

1. Administrative & Study Identification

Full Study Title: Study of Efficacy and Safety of Ribociclib (LEE011) in Combination With Topotecan and Temozolomide (TOTEM) in Pediatric Patients With Relapsed or Refractory Neuroblastoma and Other Solid Tumors **Short Title/Acronym:** Ribociclib + TOTEM in Pediatric Relapsed/Refractory Solid Tumors **Protocol Number:** CLEE011Q12101 **NCT Number:** NCT05429502 **Posting ID:** 1092181086492249338 **Sponsor Name:** Novartis **Trial Status:** TERMINATED (Study was terminated early) **Investigational Product:** Ribociclib (LEE011) in combination with Topotecan and Temozolomide (Trade name: Temodar; ATC Code: L01AX03) **Phase of Development:** Phase 1 **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the maximum tolerated dose (MTD) and/or the recommended Phase II dose (RP2D) of ribociclib in combination with topotecan and temozolomide (TOTEM) in pediatric patients. **Secondary Objectives:** To assess the safety, tolerability, pharmacokinetics, and anti-tumor activity (efficacy) of ribociclib combined with TOTEM across multiple expansion cohorts.

2.2 Background & Rationale To address the lack of effective treatments for pediatric patients with relapsed or refractory solid tumors, including neuroblastoma (NB), medulloblastoma (MB), high-grade glioma (HGG), malignant rhabdoid tumors (MRT), and rhabdomyosarcoma (RMS). Combining a CDK4/6 inhibitor (ribociclib) with standard chemotherapeutics (topotecan and temozolomide) aims to improve response rates and survival outcomes in patients who have not responded adequately to previous therapies. *Note: Due to the early termination of the trial, Phase I - Part B (multiple expansion cohorts) and Phase II (double-blind, randomized, placebo-controlled in relapsed or refractory NB) were not initiated.*

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Phase 1 dose-finding) **Blinding:** Open-label (Phase 1) **Randomization:** N/A (Phase 1)

3.2 Planned Enrollment & Duration Targeted Enrollment: 12 participants (adjusted due to early termination) **Number of Sites:**

Multicenter (~9 countries) **Estimated Study Start:** Late 2022
Estimated Primary Completion: 28-Jan-2028 (Study terminated early)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 12 months to 21 years with Lansky Play score $\geq 50\%$ (≤ 16 years) or Karnofsky performance status $\geq 50\%$ / ECOG < 3 (>16 years). 2. Histologically proven relapsed or refractory neuroblastoma (NB) or other solid tumors including medulloblastoma, high-grade glioma, malignant rhabdoid tumors, and rhabdomyosarcoma. 3. Life expectancy of ≥ 12 weeks at the time of enrollment, with adequate bone marrow, hepatic, renal, and cardiac function.

4.2 Exclusion Criteria 1. Pregnant or lactating females, or sexually active individuals unwilling to use highly effective contraception. 2. Clinically significant, uncontrolled heart disease and/or cardiac repolarization abnormality (e.g., history of QTc prolongation). 3. Prior treatment with a CDK4/6 inhibitor, allogeneic stem cell transplant within 3 months, or myeloablative therapy with autologous hematopoietic stem cell rescue within 8 weeks.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ribociclib dose escalated with topotecan and temozolomide (Temodar). Starting dose of topotecan is administered at the standard dose given to neuroblastoma patients (0.75 mg/m2/day). **Route of Administration:** Oral (Ribociclib, Temozolomide) and IV/Oral (Topotecan). **Duration of Treatment:** Up to 1 year, or longer if the disease remains stable or improves and the participant tolerates the study treatment well (subject to early trial termination).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing chemotherapy-induced nausea and myelosuppression. **Prohibited:** Medications with a known risk to prolong the QT interval, drugs mainly metabolized by CYP3A4/5 with a narrow therapeutic index, strong inducers/inhibitors of CYP3A4/5, herbal preparations, dietary supplements, and live attenuated vaccines within 4 weeks.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Tumor Status, Negative Serum Pregnancy Test (within 7 days of first dose) | | **Baseline/Treatment** | Day 0 to 1 Year | Ribociclib + TOTEM administration, Safety Labs, ECG (QTc

monitoring), Efficacy Assessments (~29 visits planned) | | **End of Study** | Follow-up | Final Vital Signs, Exit Interview, Safety/Contraception monitoring for 6 months post-treatment |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Determination of the Maximum Tolerated Dose (MTD) and/or Recommended Phase II Dose (RP2D) of ribociclib in combination with topotecan and temozolomide.

Measurement Method: Clinical assessment of Dose-Limiting Toxicities (DLTs) during Phase 1 Part A. Percentage of participants with DLTs in Cycle 1.

7.2 Secondary & Exploratory Endpoints Safety and anti-tumor activity (Overall Response Rate) of ribociclib + TOTEM in tumor-specific expansion cohorts (Phase 1b - not initiated). Pharmacokinetics (PK) of ribociclib (AUC, Cmax) and biomarker evaluations.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for acute toxicities, specifically focusing on hematologic suppression (bone marrow) and cardiac repolarization (QTc prolongation) via regular ECGs. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness, particularly those resulting in treatment interruption or discontinuation. **Data Monitoring Committee (DMC):** Internal and external safety review procedures utilized throughout the active duration of the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for Phase 1 dose-escalation.

Sample Size Justification: The targeted enrollment was capped at 12 participants due to the early termination of the study. (The original planned target was $N=231$ covering all phase 1/2 cohorts). **Handling of Missing Data:** Per protocol specification, evaluated in the context of the early termination framework.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki up until termination. **Monitoring Plan:** High-frequency onsite monitoring (approx. 29 visits per participant planned). **Audit Trail:** eCRF locking, query generation, and resolution via validated electronic data capture (EDC) systems.

11. Study Identifiers & Governance

Primary ID: CLEE011Q12101 **Registry Number:** NCT05429502 **Posting ID:** 1092181086492249338 **Sponsor/Lead Organization:** Novartis **Study Title:** Study of Efficacy and Safety of Ribociclib (LEE011) in Combination With Topotecan and Temozolomide (TOTEM) in Pediatric Patients With Relapsed or Refractory Neuroblastoma and Other Solid Tumors **Official Regulatory Title:** Phase I/II multicentre study to assess efficacy and safety of ribociclib (LEE011) in combination with topotecan and temozolomide (TOTEM) in paediatric patients with relapsed or refractory neuroblastoma and other solid tumours

12. Clinical Framework (Oncology Focus)

Primary Disease Areas: Relapsed or Refractory Neuroblastoma **Disease Classifications (Neuroblastoma):** * **MeSH Code:** D009447 * **HPO Code:** HP:0003006 * **SNOMED ID:** 87364003 **Diagnosis Threshold:** Histologically proven recurrent or refractory disease per appropriate staging criteria (e.g., INSS for Neuroblastoma) **Medical Methodology:** CDK4/6 Inhibition + Standard Chemotherapy (TOTEM) **Optimization Target:** Maximum Tolerated Dose (MTD) and objective tumor response

13. Study Procedures & Methodology

Management Pathway: Sequential dose-escalation (Phase 1a). Subsequent cohorts were not initiated due to early termination. **Education Protocol (HCP):** Protocol-specific training on dose-limiting toxicity identification and stringent QTc monitoring. **Education Protocol (Patient):** Contraception counseling and medication administration scheduling. **Quality Audit Mechanism:** Continuous monitoring of dose-limiting toxicities and protocol adherence.

14. Observation & Follow-Up Schedule

Total Duration: Up to 1 year of active treatment planned, plus 6 months post-treatment follow-up. **Onsite Visit Cadence:** Approximately 29 visits designed for the active study period. **Remote Monitoring Frequency:** Intermittent remote assessments as standard of care allowed. **Checklist Review Interval:** Continuous assessment prior to each dosing cycle.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |